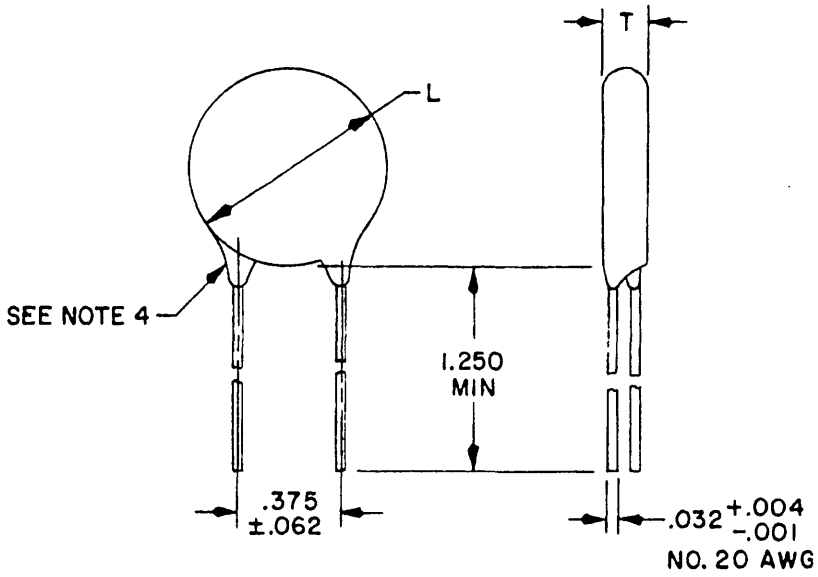
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, CERAMIC DIELECTRIC (GENERAL PURPOSE), STYLES CK64, CK65, CK66, CK67, CK68, AND CK69

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the capacitors described herein shall consist of this document and the latest issue of Specification MIL-C-11015.



INCHES	MM
.001	.03
.004	.10
.032	.81
.062	1.57
.375	9.53
1.250	31.75

Style	Dimensions	
	L ± .040 (1.02)	T ± .040 (1.02)
CK64	.730 (18.54)	.167 (4.24)
CK65	.790 (20.07)	.167 (4.24)
CK66	.890 (22.61)	.167 (4.24)
CK67	.950 (24.13)	.167 (4.24)
CK68	1.050 (26.67)	.330 (8.38)
CK69	1.110 (28.19)	.330 (8.38)

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are in parentheses.
3. Metric equivalents are given for general information only and are based upon 1.00 inch = 25.4 mm.
4. Insulating coating shall not extend more than .125 (3.18 mm) along lead wires measured from a tangent to the coating surface drawn perpendicular to the lead wires.

FIGURE 1. Dimensions and configuration.

(E) denotes changes.

FSC 5910

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TABLE I. Styles CK64, CK65, CK66, CK67, CK68, and CK69 characteristics.

Ⓔ Type designation 1/	Rated voltage	Rated temperature and voltage-temperature limits	Capacitance	Capacitance tolerance
	Volts, dc		pF	
CK64AW511M-	1,600	AW	510	M
CK64AW681M-	1,600	AW	680	M
CK64AW821M-	1,600	AW	820	M
CK64AW102M-	1,600	AW	1,000	M
CK64AW152M-	1,600	AW	1,500	M
CK64AW222M-	1,600	AW	2,200	M
CK64AW332M-	1,600	AW	3,300	M
CK64AW392M-	1,600	AW	3,900	M
CK65AW472M-	1,600	AW	4,700	M
CK66AW562M-	1,600	AW	5,600	M
CK67AW682M-	1,600	AW	6,800	M
CK67AW752M-	1,600	AW	7,500	M
CK68AW103M-	1,600	AW	10,000	M
CK69AW153M-	1,600	AW	15,000	M

Ⓔ 1/ The complete type designation will include the letter "E" to indicate an epoxy coated capacitor (when applicable) or "-" will be deleted for wax impregnated case.

REQUIREMENTS

Design and construction:

Dimensions and configuration - See figure 1.

Ⓔ Case type - Disk, wax impregnated or epoxy coated (E).

Capacitance value - See table I.

Capacitance tolerance - ±20 percent (M).

Rated temperature - -55° to +85°C.

Dielectric withstanding voltage:

Dielectric (DWV): Method 301 of MIL-STD-202.

Test voltage - 250 percent of rated voltage.

Body insulation: In accordance with MIL-C-11015.

Test potential - 3,000 volts dc.

Barometric pressure (reduced): Method 105 of MIL-STD-202, condition B (50,000 ft).

Test potential - 2,000 volts dc.

Insulation resistance (IR): Method 302 of MIL-STD-202, condition B. 200,000 megohms, minimum.

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Capacitance (Cap.): Method 305 of MIL-STD-202.

Test frequency - 1 kHz ±100 Hz.

Test voltage - 1.0 ±0.2 volt rms.

Dissipation factor (DF): 1.5 percent, maximum.

Shock (specified pulse): Method 213 of MIL-STD-202, condition I (100 G's).

Vibration, high frequency: Method 204 of MIL-STD-202, condition D (20 G's).

(E) Thermal shock and immersion:

Thermal shock: Method 107 of MIL-STD-202, condition A.

Immersion: Method 104 of MIL-STD-202, condition B.

DWV - 250 percent of rated voltage.

IR - 150,000 megohms, minimum.

Salt spray (corrosion): Not applicable.

Terminal strength: In accordance with MIL-C-11015, 5 pounds applied.

Moisture resistance: Method 106 of MIL-STD-202.

DWV - 250 percent of rated voltage.

IR - 150,000 megohms, minimum.

Cap. - Within tolerance of table I value.

Solderability: Method 208 of MIL-STD-202. 2 terminals.

Resistance to soldering heat: Method 210 of MIL-STD-202, condition B.

IR - 200,000 megohms, minimum.

ΔCap. - ±5 percent of initial measurement.

ΔDF - 0.5 percent, maximum.

Voltage-temperature limits: In accordance with MIL-C-11015.

Life (at elevated ambient temperature): Method 108 of MIL-STD-202. 2,000 hours (for qualification); 250 hours (performance check); 1,750 hours (continuation test).

Test potential - 200 percent of rated voltage.

DWV - 250 percent of rated voltage (at 25°C).

IR - 100,000 megohms, minimum (at 85°C and 25°C).

Cap. - Within tolerance of table I value (at 25°C).

DF - 1.5 percent, maximum (at 25°C).

Marking: In accordance with MIL-C-11015.

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Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - AS, OS, SH
Air Force - 11, 17
DLA - ES

User activities:
Navy - MC
Air Force - 19

Preparing activity:
Navy - EC

Agent:
DLA - ES

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DOCUMENT IDENTIFIER AND TITLE
MIL-C-11015/21E

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

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DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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